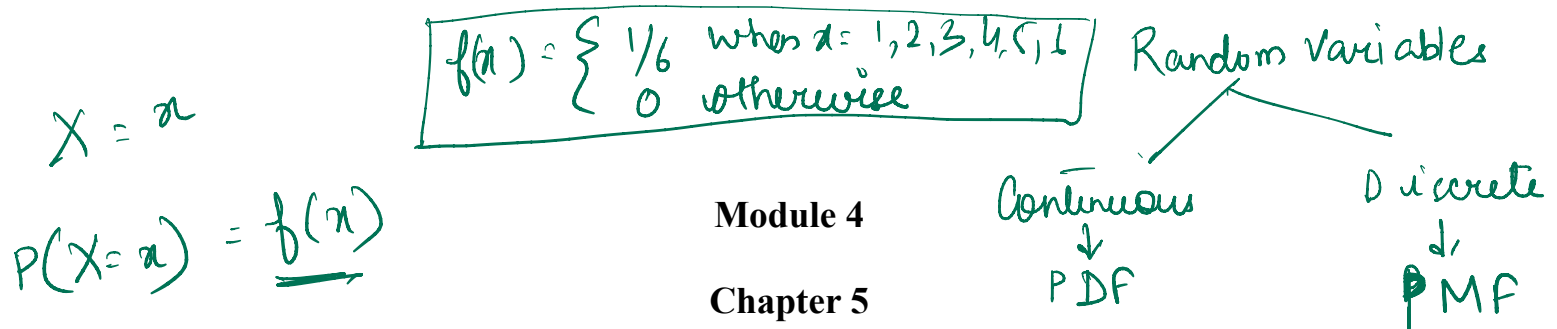




Probability Distribution and Variance

Revisiting the Random Variable

A Random Variable is a variable that takes on different values determined by chance. There are two types of Random Variables:

- **Discrete Random Variable:** A random variable that takes on a finite or countably infinite number of values. E.g. Results of a fair die roll
- **Continuous Random Variable:** A random variable that takes on a noncountable infinite number of values. E.g. Weight of a randomly chosen person

Probability Distribution Functions

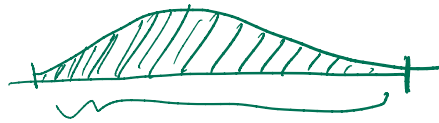
A **probability distribution function** is a mathematical **function** that provides the **distribution of probabilities for the possible outcomes of the random variable, X**. It is typically denoted as $f(x)$.

There are two classes of probability distribution functions:

Probability Mass Functions (PMF): The probability distribution for a **discrete random variable** is usually called a PMF. If X is discrete, then $f(x) = P(X = x)$. In other words, the **PMF** for a constant, x , is the probability that the random variable **X is equal to x** . The PMF can be in the form of an equation, or it can be in the form of a table.

Properties of PMF:

- $f(x) > 0$ for x in sample space and 0 otherwise.

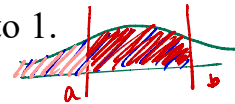
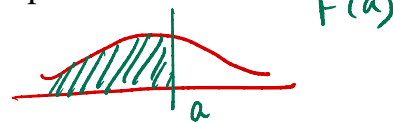


- $\sum_x f(x) = 1$. In other words, the sum of the probabilities of all possible outcomes of a given experiment equals to 1.

Probability Density Functions (PDF): the probability distribution function of a *continuous random variable* X is called a PDF. If X is continuous, $P(X = x) = 0$ for any given value x . Unlike the discrete random variables, the **PDF** of a *continuous* random variable **does not** equal to $P(X = x)$. To find probabilities over an interval, such as $P(a < X < b)$, using the PDF would require calculus.

Properties of a PDF:

- $f(x) > 0$ for x in sample space and 0 otherwise.
- The **area under the curve** is equal to 1.



The probability of a random variable being less than or equal to a given value is calculated using another probability function called the **cumulative distribution function**.

$$P(a \leq X \leq b) = F(b) - F(a) = \int_a^b f(x) dx = F(b)$$

Cumulative Distribution Function (CDF): A **cumulative distribution function** (CDF), usually denoted $F(x)$, is a function that gives the probability that the random variable, X , is **less than or equal to the value x** .

$$F(x) = P(X \leq x)$$

** The definition of the cumulative distribution function is the same for a discrete random variable or a continuous random variable**

Let's look at an example

Consider the data set with the values: 0,1,2,3,4. If X is a random variable of a random draw from these values, and if we assume the probabilities of all the outcomes were the same, the **PMF** could be displayed in **function** form or a **table**. As a function, the PMF would look like:

$$f(x) = \begin{cases} \frac{1}{5} & \text{if } x = 0, 1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

As a table, the **PMF** would look like:

<i>x</i>	<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
<i>f(x)</i>	<i>1/5</i>	<i>1/5</i>	<i>1/5</i>	<i>1/5</i>	<i>1/5</i>

We can also find **CDF** using the above information:

Let's find $F(2)$:

$$F(2) = P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{1}{5} + \frac{1}{5} + \frac{1}{5} = \frac{3}{5}$$

Variance of a Random Variable

Consider the following PDF of the random variable representing the sum of two 6-sided dice:

x	2	3	4	5	6	7	8	9	10	11	12
f(x)	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

It's clear that **7** is the **most common** single result, and that there's a **symmetric drop-off** in probability, where **2 and 12 are least common**. It seems like we should feel more confident that we'll get a value in the middle somewhere. This aspect of the distribution is captured by its variance, a number that, like the expectation, characterizes something about what we should expect from this distribution in the long run.

Variance captures something about what to expect for **how often we'll be far from expectation**. Going back to the lottery example, if I hold the ten bucks in my pocket, I'll always get what I expect: the ten bucks will still be in my pocket. But if

I buy the lottery ticket, I'll never actually get my expectation as a result – what actually happens could be far from the expected value. That's variance.

The variance of a *discrete random variable* is given by:

$$\sigma^2 = \text{Var}(X) = \sum (x_i - \mu)^2 f(x_i)$$

*****We take each value of x, subtract the expected value, square that value and multiply that value by its probability. Then sum all of those values*****

The ***Standard Deviation (σ)*** of a random variable is the ***square root of Variance***.

Let's take an example

Calculate the variance of a 6-sided die roll, knowing that the expectation is 3.5.

We know,

$$\sigma^2 = \frac{1}{6} \times \{(1 - 3.5)^2 + (2 - 3.5)^2 + (3 - 3.5)^2 + (4 - 3.5)^2 + (5 - 3.5)^2 + (6 - 3.5)^2\} = 2.916$$

Properties of Variance and Expectation

Both expectation and variance have nice properties that make it easier to compute them for complex distributions when those distributions are the result of summing simpler random variables:

▪ $E(c) = c$	▪ $\text{Var}(c) = 0$
▪ $E(cX) = c E(X)$	▪ $\text{Var}(cX) = c^2 \text{Var}(X)$

▪ $E(X + c) = E(X) + c$	▪ $Var(X + c) = Var(X)$
▪ $E(X + Y) = E(X) + E(Y)$	▪ $Var(X + Y) = Var(X) + Var(Y)$ <i>Note: X and Y need to be independent</i>

Let's look at an example

$$Var[X + X] = Var[X] + Var[X]$$

$$Var[2X] = 2^2 Var[X]$$

The variance of the *sum of two dice* will be *twice* the variance of a *single six sided die roll* which we calculated in the last example: $2 \times 2.916 = 5.832$ (Using the property: $Var(X + Y) = Var(X) + Var(Y)$. Here my new random variable is $(X + X)$)

However, if a game said, “*Double your six-sided die roll*”. The variance in that case will be: $2^2 \times 2.916 = 11.664$ (Using the property: $Var(cX) = c^2 Var(X)$.) Since, here my new random variable is $2X$.

Let's Practice

- Given the following function:

$$f(x) = \begin{cases} \frac{3}{37760} x^2 (20 - x) & \text{if } 2 \leq x \leq 18 \\ 0 & \text{otherwise} \end{cases}$$

- Show that $f(x)$ is a probability density function. (Hint: the total area under the curve should be =1)
 - Find $P(X \leq 7)$ $f(7)$
 - Find $P(X \geq 7)$
 - Find $P(3 \leq X \leq 14)$
- Suppose X and Y are independent random variables and $Var[X] = 3$ and $Var[Y] = 5$. Calculate the following:
 - $Var[3X + 4]$
 - $Var[X + 3Y]$
 - $Var[X - Y]$

- Let X and Y be two independent random variables. Suppose we know that $\text{Var}[2X - Y] = 6$ and $\text{Var}[X + 2Y] = 9$. Calculate $\text{Var}[X]$ and $\text{Var}[Y]$.
- Let X be a discrete random variable with the given pmf:

$$f(x) = \begin{cases} 0.2 & \text{when } x = 0, 1 \\ 0.3 & \text{when } x = 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

We define a random variable $Y = X(X - 1)(X - 2)$. Find the pmf of Y .

1. a) To show that $f(x)$ is a valid PDF, we need to show that the area under the curve = 1

i.e. $F(\infty) = \int_{-\infty}^{\infty} \frac{3}{7760} x^2 (20 - x) dx = 1$

$$F(\infty) = \int_{-\infty}^2 0 dx + \int_2^{18} \frac{3 x^2}{7760} (20 - x) dx + \int_{18}^{\infty} 0 dx$$

$$= 0 + \left[\frac{3}{7760} \left(\frac{20x^3}{3} - \frac{x^4}{4} \right) \right]_2^{18} + 0$$

$$= 0 + 1 + 0$$

$$b) \quad P(X \leq 7) = F(7) = \int_{-\infty}^7 f(x) dx = \int_2^7 f(x) dx$$

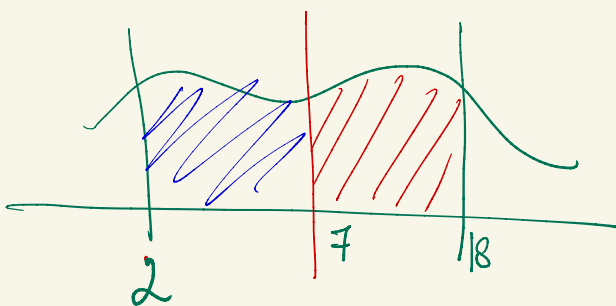
$$= \int_2^7 \frac{3}{37760} x^2 (20-x) dx = 0.130665$$

$$c) \quad P(7 \leq X \leq 18) = \int_7^{18} f(x) dx = \int_7^{18} \frac{3 x^2 (20-x)}{37760} dx$$

$$= 0.869935$$

$$\text{OR} \quad (1 - P(X \leq 7))$$

why?



$$\begin{aligned}
 d) \quad P(3 \leq X \leq 14) &= \int_3^{14} f(x) dx = \int_3^{14} \frac{3x^2(20-x)}{37760} dx \\
 &= 0.677668
 \end{aligned}$$

2. Given,

$$\text{Var}[X] = 3$$

$$\text{Var}[Y] = 5$$

(X and Y are independent)

$$\begin{aligned}
 a) \quad \text{Var}[3X+4] &= \text{Var}[3X] + 0 \\
 &= 3^2 \text{Var}[X] = 3 \times 3 = 3^2 = 27
 \end{aligned}$$

$$\begin{aligned}
 b) \quad \text{Var}[X+3Y] &= \text{Var}[X] + 3^2 \text{Var}[Y] \\
 &= 3 + 9 \times 5 = 3 + 45 = 48
 \end{aligned}$$

$$\begin{aligned}
 c) \quad \text{Var}[X-Y] &= \text{Var}[X] + \text{Var}[-Y] \\
 &= \text{Var}[X] + \text{Var}[(-1)Y] \\
 &= \text{Var}[X] + (-1)^2 \text{Var}[Y] \\
 &= \text{Var}[X] + \text{Var}[Y] \\
 &= 3 + 5 = 8
 \end{aligned}$$

3. Given $\text{Var} [2x - y] = 6$

$$\text{Var} [x + 2y] = 9$$

$$2^2 \text{Var} [x] + (-1)^2 \text{Var} [y] = 6$$

$$\Rightarrow 4 \text{Var} [x] + \text{Var} [y] = 6 \quad \dots (i)$$

Similarly,

$$\text{Var} [x] + 2^2 \text{Var} [y] = 9$$

$$\text{Var} [x] + 4 \text{Var} [y] = 9 \quad \dots (ii)$$

Eq (i) and (ii) are linear equations

Solving them, we get:

$$\text{Var} [y] = 2$$

$$\text{Var} [x] = 1$$

4.

given,

$$f(x) = \begin{cases} 0.2 & \text{for } x = 0, 1 \\ 0.3 & \text{for } x = 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

$$Y = X(X-1)(X-2)$$

possible values of Y : $X(X-1)(X-2)$ for
 $x \in \{0, 1, 2, 3\}$

$\therefore$ Possible values of Y = $\{0, 6\}$
 (also called Range of Y)

$$\begin{aligned} \therefore P(Y=0) &= P(X=1) \text{ or } P(X=2) \text{ or } P(X=0) \\ &= 0.2 + 0.3 + 0.2 \\ &= 0.7 \end{aligned}$$

$$P(Y=6) = P(X=3) = 0.3$$

$\therefore$ PMF of Y :

$$f(y) = \begin{cases} 0.7 & \text{when } y = 0 \\ 0.3 & \text{when } y = 6 \\ 0 & \text{otherwise} \end{cases}$$